C3 Bonding – Knowledge Organiser

AQA A-level Chemistry • Set 1: Types of Bonding and Ions **Papers 1 & 2**

#	Question	Answer
1	Ionic bonding	Electrostatic attraction between oppositely charged ions
2	Covalent bond	Shared pair of electrons
3	Dative covalent (coordinate) bond	Shared pair of electrons both from one atom
4	Metallic bonding	Attraction between positive ions and delocalised electrons
5	Example of a species with a dative bond	NH_4^+
6	Formula of the sulfate ion	SO_4^{2-}
7	Formula of the nitrate ion	NO_3^-
8	Formula of the carbonate ion	CO_3^{2-}
9	Formula of the ammonium ion	NH_4^+
10	Formula of the hydroxide ion	OH^-

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AQA A-level Chemistry • Set 2: Structures **Papers 1 & 2**

#	Question	Answer
1	Structure of sodium chloride	Giant ionic lattice
2	Structure of diamond	Giant covalent (macromolecular)
3	Structure of graphite	Giant covalent, in layers
4	Structure of iodine	Simple molecular
5	Structure of magnesium	Giant metallic lattice
6	Bonds per carbon atom in diamond	4
7	Bonds per carbon atom in graphite	3
8	Why graphite conducts electricity	Delocalised electrons
9	Why ionic compounds conduct when molten	Ions free to move
10	Why metals are malleable	Layers of ions slide

C3 Bonding – Knowledge Organiser

AQA A-level Chemistry • Set 3: Shapes of Molecules Papers 1 & 2

#	Question	Answer
1	2 bonding pairs, 0 lone pairs	Linear, 180°
2	3 bonding pairs, 0 lone pairs	Trigonal planar, 120°
3	4 bonding pairs, 0 lone pairs	Tetrahedral, 109.5°
4	3 bonding pairs, 1 lone pair	Trigonal pyramidal, 107°
5	2 bonding pairs, 2 lone pairs	Bent, 104.5°
6	5 bonding pairs, 0 lone pairs	Trigonal bipyramidal, 120° and 90°
7	6 bonding pairs, 0 lone pairs	Octahedral, 90°
8	4 bonding pairs, 2 lone pairs	Square planar, 90°
9	Bond angle reduction per lone pair	About 2.5°
10	Order of electron pair repulsion	LP–LP > LP–BP > BP–BP

C3 Bonding – Knowledge Organiser

AQA A-level Chemistry • Set 4: Electronegativity and Intermolecular Forces **Papers 1 & 2**

#	Question	Answer
1	Electronegativity	Power of an atom to attract the pair of electrons in a covalent bond
2	Most electronegative element	Fluorine
3	Trend in electronegativity across a period	Increases
4	Trend in electronegativity down a group	Decreases
5	Polar bond	Unequal sharing of electrons
6	Three types of intermolecular force	Induced dipole–dipole (van der Waals), permanent dipole–dipole, hydrogen bonding
7	Atoms H must bond to for hydrogen bonding	N, O or F
8	Strongest intermolecular force	Hydrogen bonding
9	Why ice is less dense than water	Open lattice held by hydrogen bonds
10	Factor increasing van der Waals strength	More electrons

C3 Bonding – Knowledge Organiser

AQA A-level Chemistry • Set 5: Named Shapes and Polarity **Papers 1 & 2**

#	Question	Answer
1	Shape of CH_4	Tetrahedral
2	Shape of BF_3	Trigonal planar
3	Shape of NH_4^+	Tetrahedral
4	Shape of PCl_5	Trigonal bipyramidal
5	Shape of SF_6	Octahedral
6	Shape of XeF_4	Square planar
7	Bond angle in H_2O	104.5°
8	Bond angle in NH_3	107°
9	Why CO_2 is non-polar	Symmetrical, so dipoles cancel
10	Example of a non-polar molecule with polar bonds	CCl_4 (or CO_2)